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Original article

Emotion-driven music recommendation system based on fully convolutional recurrent attention networks and collaborative filtering

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ABSTRACT

With the rapid growth of online music platforms, personalized music recommendation has become a critical task. However, existing methods often struggle to capture the intricate emotional and contextual characteristics of music, resulting in suboptimal user experiences. To address these limitations, we propose a hybrid recommendation system that integrates collaborative filtering, music attribute modeling, and a FCRA. The FCRA extracts key emotional features, including tempo, pitch variation, spectral contrast, harmonic progression, and rhythm patterns, which correspond to affective states such as happiness, sadness, calmness, and intensity. The hybrid design leverages both user interaction history and music content to enhance recommendation accuracy. Experimental results on the MSD and Last.Fm 360k datasets show that our model achieves a Precision@10 of 0.902 and 0.885, Recall@10 of 0.832 and 0.815, and NDCG@10 of 0.855 and 0.84, respectively, outperforming all baseline methods. Furthermore, the system performs robustly in cold-start scenarios, highlighting its adaptability across different recommendation contexts. This research provides a novel approach to addressing the challenges of emotion-driven music recommendation, contributing to improved user satisfaction and better alignment with individual preferences and moods.

1. Introduction

With the rise of digital music streaming, online platforms such as Spotify and Apple Music have transformed the way users access and consume music [1,2]. With global streaming users increasing rapidly, platforms like Spotify and Apple Music dominate the industry, providing vast music libraries but also leading to information overload [3,4]. Recommendation systems help address this issue, yet traditional collaborative filtering and content-based methods struggle to capture the emotional context of music and evolving user preferences, limiting their personalization capabilities [5,6]. As a result, users often experience mismatches in recommendations, reducing engagement and satisfaction. To overcome these challenges, designing an efficient and adaptive music recommendation system that incorporates emotional understanding and dynamic personalization is crucial for enhancing user experience and industry development.

Traditional recommendation methods, such as collaborative filtering and content-based approaches, primarily rely on user interaction data or explicit item attributes to generate recommendations [7,8]. While these techniques perform well in data-rich scenarios, they suffer from limitations such as the cold-start problem and an inability to capture users' evolving emotional states [9,10]. Hybrid recommendation models attempt to combine multiple techniques to improve performance [11,12], but they often depend on static user preferences and struggle to adapt to real-time emotional changes. Additionally, existing models focus primarily on explicit user behaviors while overlooking the deeper emotional and contextual factors that influence music choices, leading to recommendations that may not align with users' current moods or listening intent [13].

With the advancements in artificial intelligence, deep learning has opened new possibilities for improving recommendation systems. By leveraging automated feature extraction and non-linear representation learning, deep learning models can capture intricate relationships between user behavior and music features, significantly enhancing recommendation accuracy [14,15]. Convolutional neural networks (CNNs) effectively process high-dimensional data such as music spectrograms, extracting meaningful audio features, while recurrent neural networks (RNNs) model sequential dependencies in user behavior and dynamic changes in music preferences [16,17]. Despite these advancements, most existing research has focused on general interest-based recommendations, with limited emphasis on modeling emotional responses

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and adapting to users' dynamic preferences [18]. Moreover, music recommendation faces unique challenges, such as real-time responsiveness and recommendation diversity, which become even more critical in emotion-driven scenarios [19]. For instance, when users seek music that aligns with their immediate mood, existing systems often struggle to provide emotionally relevant recommendations in real time, leading to a disconnect between user intent and system output. These limitations highlight the need for novel approaches that integrate emotional understanding, contextual awareness, and adaptive personalization to enhance the quality and effectiveness of music recommendations.

To address the limitations of current music recommendation systems, particularly in emotion modeling and personalized user preference matching, this paper proposes a hybrid recommendation framework that integrates deep learning-based emotion recognition with traditional recommendation techniques. Unlike conventional systems that rely solely on collaborative filtering or content-based methods, our approach combines multiple strategies to enhance recommendation accuracy and personalization. The system integrates an FCRA, collaborative filtering, and a music attribute-based recommendation model. By leveraging deep learning techniques, it extracts emotional characteristics from music and combines these features with user behavior data and music attributes to create a personalized recommendation system that dynamically adapts to users' emotional needs. The FCRA network utilizes convolutional layers to capture local musical features, recurrent layers to model time-dependent patterns, and an attention mechanism to highlight key emotional aspects. Meanwhile, the collaborative filtering model analyzes historical user behavior data to identify relationships between users and items, generating initial recommendations. The music attribute model aligns emotional tags, styles, and melodic features with users' emotional preferences, proving particularly beneficial in cold-start situations where user data is sparse. By explicitly incorporating emotion modeling into the recommendation process, our system effectively matches music with users' real-time emotional states, a capability often lacking in existing systems. This approach improves recommendation precision while ensuring stronger emotional alignment, allowing the system to dynamically adapt to users' moods and preferences in real time.

Main Contributions of This Paper:

- This paper introduces the FCRA network, which effectively models music emotions by integrating convolutional layers, recurrent layers, and an attention mechanism to capture local features, temporal dependencies, and key emotional attributes.
- This paper develops a hybrid recommendation strategy that combines collaborative filtering for data-rich scenarios and music attribute-based methods for cold-start situations, enhancing adaptability and robustness.
- This work introduces an emotion matching mechanism that aligns music recommendations with users' long-term preferences and immediate emotional states, enabling dynamic and personalized recommendations.

The remainder of this paper is organized as follows: Section 2 provides an overview of related work, discussing recommendation algorithms and the role of deep learning in music recommendation. Section 3 presents the proposed hybrid recommendation framework, including the structure of the FCRA network and its integration with collaborative filtering and content-based models. Section 4 details the experimental setup and results, followed by Section 5, which provides a discussion of the method's advantages, limitations, and potential improvements. Finally, Section 6 concludes the paper.

2. Related work

2.1. Recommendation algorithms

Recommendation algorithms play a crucial role in personalized information filtering and have been widely applied in various fields, including e-commerce, news, social media, and music streaming [20,21]. Their core function is to analyze user behavior and interest preferences to generate personalized recommendations, thereby improving user experience and platform engagement. Traditional recommendation methods mainly include collaborative filtering, content-based recommendation, and hybrid strategies [22,23]. While these methods have significantly advanced recommendation systems, they still face challenges such as data sparsity, cold start problems, and limited adaptability to users' emotional needs.

Collaborative Filtering (CF) is one of the most widely used techniques in recommendation algorithms. It generates recommendations by analyzing user-item interactions [24]. User-based CF identifies users with similar preferences, while item-based CF analyzes co-occurrence relationships between items [25]. Matrix factorization techniques, such as Singular Value Decomposition (SVD) and Non-negative Matrix Factorization (NMF), further improve CF by learning latent representations of users and items [26,27]. However, CF often relies on sufficient interaction data to perform effectively, and in cold-start scenarios, where user or item data is limited, its performance may decline. Content-Based Recommendation methods generate recommendations by analyzing item attributes, making them particularly useful for cold-start problems [28]. In music recommendation, content features such as frequency spectrum, MFCCs, and lyric sentiment analysis help match user preferences [29,30]. While effective, content-based methods can sometimes lead to "over-specialization", where recommendations focus too narrowly on known preferences, potentially reducing recommendation diversity. To address the limitations of individual methods, researchers have proposed Hybrid Recommendation Strategies that combine multiple approaches to improve performance [31]. Common strategies include weighted fusion, where predictions from different models are combined, and cascaded fusion, where one method refines the output of another. For example, CF can capture user similarity, while content-based methods analyze item features to improve recommendation diversity [32]. Recently, context-aware hybrid strategies have integrated additional factors, such as time, location, and user activity, to enhance personalization [33,34]. While hybrid strategies improve recommendation quality, they also require balancing different components effectively to maintain efficiency. Despite these advancements, traditional recommendation algorithms still face difficulties in dynamically adapting to users' emotional needs [35–37]. As a result, enhancing the adaptability and emotional relevance of recommendations remains a key research challenge.

In summary, recommendation algorithms have evolved from collaborative filtering to content-based and hybrid strategies, establishing a strong foundation for modern recommender systems. However, as user preferences become more complex and data volumes continue to grow, challenges related to high-dimensional feature modeling, dynamic emotional adaptation, and real-time response persist. These limitations highlight the need for advanced approaches, particularly deep learning techniques, to further enhance recommendation performance.

2.2. Deep learning in recommendation systems

In recent years, deep learning has gained prominence in recommendation systems for its ability to extract complex features and uncover implicit user-item interactions. Unlike traditional methods that rely on explicit features and simple similarity metrics, deep learning excels in handling high-dimensional and unstructured data, enabling more accurate and personalized recommendations [38].

In user behavior modeling, deep learning techniques have proven effective in processing sequential data. They capture dynamic relationships between users' short-term interests and long-term preferences. Studies have applied Recurrent Neural Networks (RNNs) and their advanced variants, such as Long Short-Term Memory networks (LSTM) and Gated Recurrent Units (GRU), to model user behavior over time. These models excel at learning temporal dependencies, enabling the generation of dynamic and contextually relevant recommendations [39–41]. By analyzing historical user behavior, deep learning models predict future preferences more effectively than traditional collaborative filtering, which relies on static rating data. Unlike traditional methods, deep learning captures the fluidity of user preferences, significantly improving the adaptability and timeliness of recommendations.

In item feature modeling, deep learning offers clear advantages. Traditional recommendation algorithms primarily rely on explicit item features such as style, artist, or basic tags [42–44]. In contrast, deep learning methods can extract deeper representations of item characteristics. Convolutional Neural Networks (CNNs) process audio spectrograms to extract advanced features that reflect a song's emotion, melody, and rhythm [45,46]. This capability enables recommendation systems to better understand music characteristics and generate more accurate recommendations. Moreover, recent advancements in deep learning have enabled facial expression recognition to be integrated into emotion-aware music recommendation, allowing systems to dynamically adapt music selections based on real-time emotional analysis [47]. Attention mechanisms have been widely used in emotion-aware recommendation systems to enhance performance by selectively focusing on key information. In music recommendation, attention mechanisms help identify the most emotionally significant parts of a song, improving the model's ability to capture fine-grained emotional transitions. Integrating attention mechanisms with RNNs and CNNs allows systems to dynamically adjust recommendations based on users' real-time emotional states [48]. Self-attention mechanisms, such as those used in Transformer-based models, have demonstrated strong potential in learning long-range dependencies in user behavior data, further improving the effectiveness of personalized recommendations [49]. Furthermore, recent research has explored graph-based approaches for improving tag-aware music recommendations, where Graph Intention Embedding Neural Networks have shown promising results in enhancing user-item interactions by capturing intent-driven relationships [50]. These advancements highlight the crucial role of attention mechanisms in modern deep learning-based recommendation systems. The introduction of generative models has further expanded deep learning applications in recommendation systems. Generative Adversarial Networks generate recommended content that matches users' latent interests through adversarial training, improving recommendation diversity [51,52]. Some studies have applied GANs to music recommendation, addressing the novelty issue by generating user-preferred music feature representations [53,54]. Variational Autoencoders (VAEs) have also gained attention for modeling latent user behavior distributions, enabling more precise recommendations, especially in data-sparse or cold-start scenarios [55,56].

Overall, deep learning has significantly improved recommendation systems by enhancing feature extraction and relationship modeling. However, as user demands evolve and data volumes grow, further optimizations in model efficiency, feature learning, and adaptability remain essential to ensure continued advancements in recommendation technology.

3. Method

3.1. Overview of our network

This paper proposes a hybrid recommendation system architecture that combines deep learning and traditional recommendation algorithms, aiming to address the shortcomings of existing recommendation systems in music emotion modeling, dynamic adaptation, and cold-start issues. The overall architecture, as shown in Fig. 1, consists of three main modules: FCRA, Collaborative Filtering (CF) module, and Music Attribute Recommendation (MAR) module. These three modules collaborate from the perspectives of music emotion modeling, user behavior analysis, and explicit feature matching, and generate the final recommendation results through a weighted fusion strategy. The FCRA is the core module of the model, focusing on extracting deep emotion-related features from music data. Through convolutional layers, it captures local features such as melody and rhythm, and combines recurrent layers to model temporal dynamics. Additionally, the attention mechanism dynamically enhances emotionally relevant segments, improving the system's ability to model music-driven emotions more effectively. The extracted features are transformed into an emotion-aware embedding space, which is subsequently integrated into the recommendation framework.

The Collaborative Filtering (CF) module extracts personalized interest preferences based on user historical behavior data, such as ratings and play history. Using matrix factorization methods, it models the implicit associations between users and items, generating recommendations aligned with user interests. However, CF has limited performance in cold-start or data-sparse scenarios and requires assistance from other modules to enhance recommendation effectiveness. Therefore, the Music Attribute Recommendation (MAR) module directly matches music's explicit features, such as style, emotion tags, and melody characteristics, with user preferences, providing preliminary recommendations in cold-start scenarios. This module compensates for the limitations of CF in data-sparse situations, adding robustness to the recommendation system. The final recommendation is obtained through a dynamically weighted fusion strategy, which adaptively balances the contributions of the FCRA, CF, and MAR modules based on user behavior density and the availability of contextual data. Compared to static weighting approaches, this dynamic mechanism enhances recommendation flexibility and responsiveness.

By effectively integrating deep learning with traditional recommendation methods, this architecture leverages the strengths of each module to enhance both personalization and adaptability. The FCRA module demonstrates powerful capabilities in music emotion modeling, the CF module provides high-quality personalized recommendations when user behavior data is rich, and the MAR module offers strong supplementary support in cold-start scenarios. The introduction of an adaptive weighting strategy ensures that the system maintains a balance between recommendation accuracy, emotional relevance, and real-time applicability. Fig. 1 displays the core modules and their functional relationships, providing a clear technical framework for the following sections.

3.2. Fully convolutional-recurrent attention network

The FCRA is a hybrid deep learning architecture that combines convolutional neural networks (CNN), bidirectional gated recurrent units (Bi-GRU), and an attention mechanism to extract deep emotional features from music data. By leveraging the representation power of these components, FCRA is capable of modeling both local musical features, such as melody and rhythm, and the dynamic changes of emotions over time. This provides a robust foundation for personalized and emotion-aware music recommendation systems.

The design of FCRA allows it to simultaneously capture spatial features (e.g., frequency distributions, local changes) and temporal dependencies (e.g., emotional fluctuations) from music data. This is particularly crucial in music recommendation, where emotions are not only influenced by local features like pitch and rhythm but also depend on the overall temporal progression of the piece. By integrating convolutional, recurrent, and attention mechanisms, FCRA addresses the limitations of traditional methods in high-dimensional feature modeling and dynamic emotional adaptation, making it highly effective in emotion-driven recommendation scenarios.

The overall architecture of FCRA consists of three key modules: convolutional layers, bidirectional recurrent layers, and an attention layer. The input audio data is first processed into Mel spectrograms, which are two-dimensional representations capturing the energy distribution of audio signals across time and frequency. As shown in Fig. 2, this

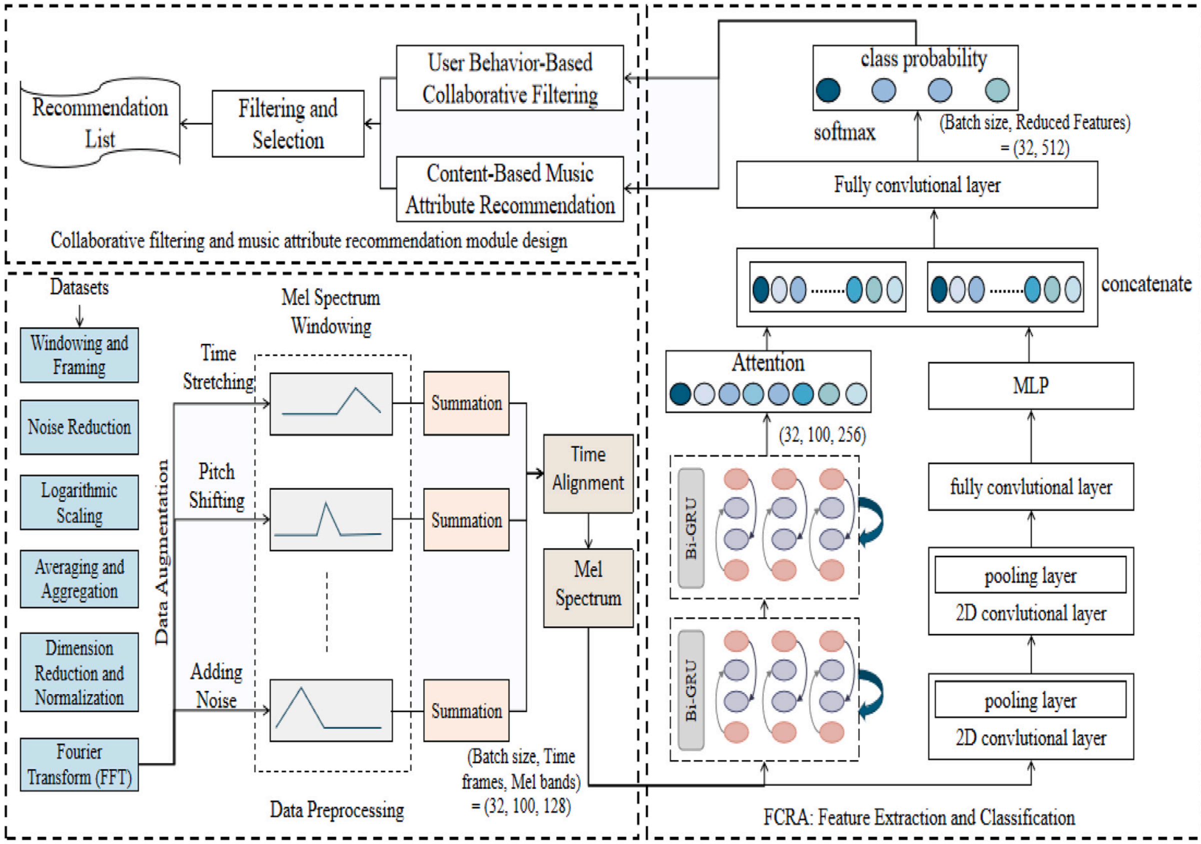

Fig. 1. The overall architecture of the proposed hybrid recommendation system, integrating collaborative filtering and music attribute recommendation modules. The preprocessing pipeline generates Mel spectrograms from audio data, which are subsequently processed through the FCRA to extract emotional and contextual features, enhancing recommendation accuracy and personalization.

architecture seamlessly integrates these components to effectively process music data for personalized recommendations. The convolutional layers apply sliding filters to extract local patterns, such as melodic fragments and rhythm changes, while pooling layers reduce the feature dimensions to retain the most important information. Subsequently, the bidirectional Bi-GRU module models both forward and backward temporal dynamics of the extracted features, effectively capturing long-term dependencies through its gating mechanism. Finally, an attention mechanism dynamically focuses on the most emotionally relevant segments of the music sequence, generating a final context vector used for recommendation tasks.

The mathematical workflow of FCRA can be described as follows.

The convolutional layers extract local features from the Mel spectrogram:

$$X_c = \mathrm{Conv}(X) \otimes W_c + b_c \tag{1}$$

where X represents the input Mel spectrogram, W_c denotes the convolutional filter weights, b_c is the bias term, and $\otimes$ indicates the convolution operation.

The output of the convolutional layers is passed to the Bi-GRU module for temporal modeling:

$$H_t = \text{Bi-GRU}(H_{t-1}, X_c) \tag{2}$$

where H_t is the hidden state at time t, H_{t-1} is the hidden state at time $t-1$, and X_c is the input sequence from the convolutional layer.

The attention mechanism computes the attention energy for each time step:

$$e_t = v_a^{\top} \tanh(W_a H_t + b_a) \tag{3}$$

where e_t represents the attention energy for time step t, v_a is the attention vector, W_a is the attention weight matrix, and b_a is the bias term.

The attention weights are then calculated using a softmax function:

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{T} \exp(e_k)} \tag{4}$$

where α_t is the attention weight for time step t.

The context vector is computed as the weighted sum of all hidden states:

$$C = \sum_{t=1}^{T} \alpha_t H_t \tag{5}$$

where C is the context vector, summarizing the time-series features with attention weights.

The context vector is transformed into an emotion embedding through a fully connected layer:

$$E = W_e C + b_e \tag{6}$$

where E is the emotion embedding, W_e is the weight matrix of the fully connected layer, and b_e is the bias term.

Finally, the emotion embedding is combined with user preferences to generate a recommendation score:

$$R = \mathrm{Sim}(U, E) \tag{7}$$

where R represents the recommendation score, U is the user preference embedding, E is the music emotion embedding, and Sim is a similarity function, such as cosine similarity.

Unlike previous models that rely primarily on static audio features or explicit user feedback, FCRA dynamically adapts to emotional shifts in music by integrating hierarchical feature representations. The attention mechanism ensures that both dominant patterns and subtle emotional variations are effectively captured, enhancing the model's adaptability. The Bi-GRU module further strengthens the system by modeling

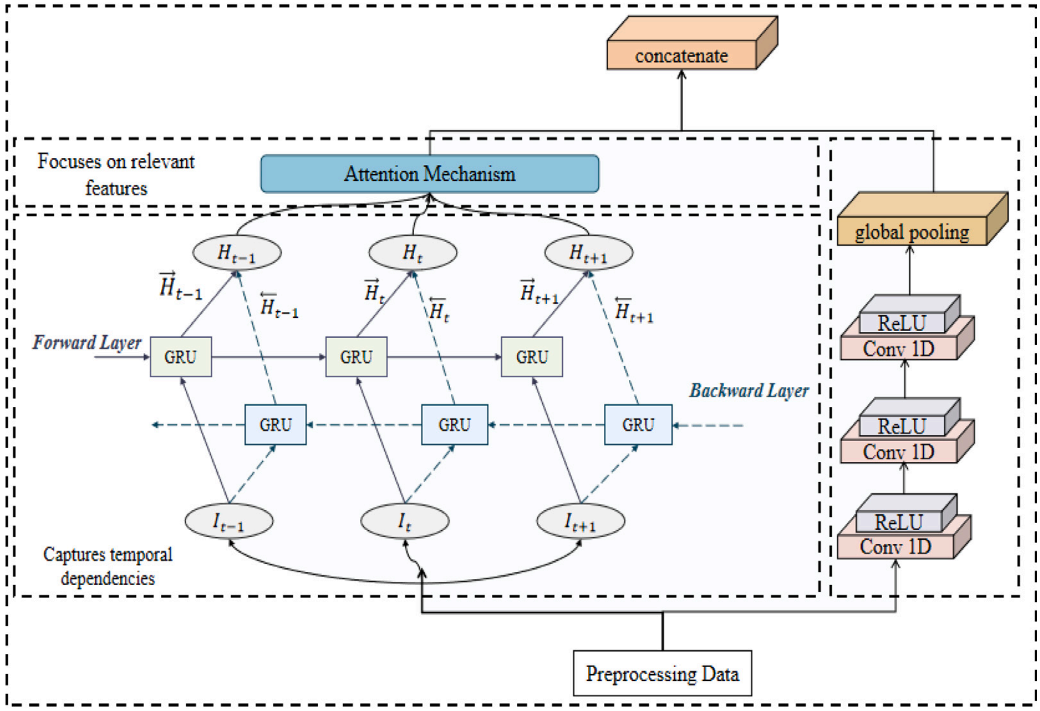

Fig. 2. Architecture of the FCRA Network.

long-range dependencies, which is crucial for capturing the evolving emotional state of a musical piece. These enhancements improve FCRA's ability to handle diverse music genres and varying emotional intensities. FCRA integrates the strengths of convolutional, recurrent, and attention mechanisms to achieve superior performance in music emotion modeling. The convolutional layers capture local patterns, the Bi-GRU layers handle temporal dynamics, and the attention mechanism enhances the model's ability to focus on emotionally significant segments. This modular design ensures efficient extraction of emotional features and dynamic adaptation to user needs. Additionally, FCRA employs an adaptive feature selection mechanism that dynamically adjusts feature weights based on the user's historical preferences and real-time emotional state. This improves recommendation precision and enables the system to respond flexibly to mood variations, making it well-suited for real-world music streaming applications. By generating emotion embeddings, FCRA provides a strong foundation for emotion-aware recommendations, allowing the system to deliver high-quality personalized results in various scenarios. Whether for initial recommendations in cold-start situations or real-time emotional adaptation, FCRA effectively enhances user experience and opens new directions for integrating emotion modeling into recommendation systems.

3.3. Collaborative filtering and music attribute recommendation module design

The hybrid recommendation system proposed in this paper integrates collaborative filtering and music attribute recommendation modules. These two modules complement each other, addressing user preference modeling in scenarios with abundant interaction data and handling cold-start scenarios where user history is insufficient. The collaborative filtering module identifies latent preferences based on user behavior data, while the music attribute recommendation module directly matches user needs with explicit attributes of music. The overall structure and workflow of this hybrid recommendation system are illustrated in Fig. 3, providing a clear representation of how the two modules interact to ensure efficient and accurate personalized recommendations in various contexts.

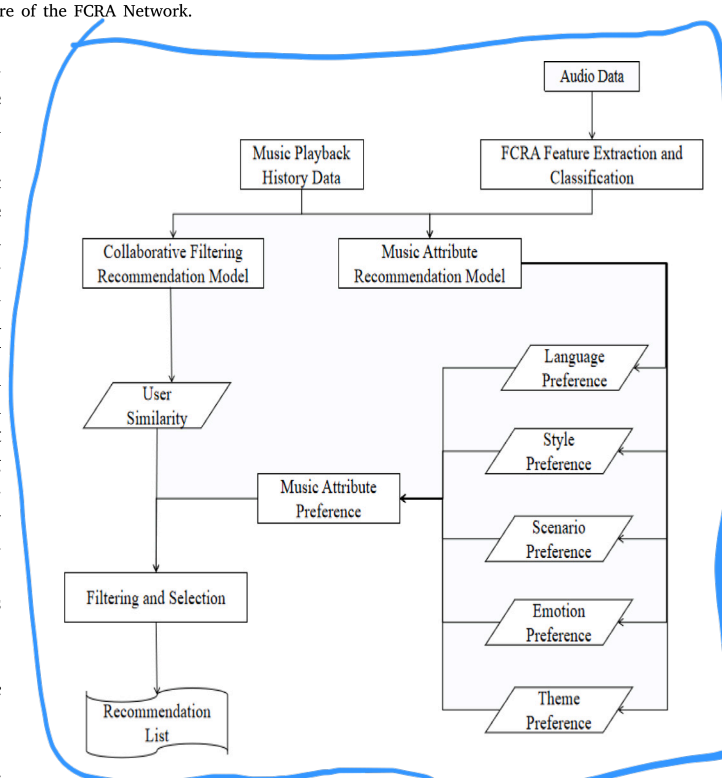

Fig. 3. Schematic representation of the proposed hybrid recommendation model. The model combines collaborative filtering based on user playback history with music attribute preferences, including language, style, scenario, emotion, and theme, to generate a personalized recommendation list.

3.3.1. Collaborative filtering module design

The collaborative filtering module analyzes interaction data, such as user ratings and playback histories, to model latent relationships between users and items. Its fundamental principle is to predict a user's preference for an unseen item based on the preferences of similar users or items. The core prediction model for collaborative filtering is expressed as:

$$R_{u,i} = \mu + b_u + b_i + Q_i^\top P_u \tag{8}$$

where $R_{u,i}$ represents the predicted rating of user u for item i, μ is the global average rating, b_u and b_i are user and item biases, and P_u and Q_i are the latent feature vectors of the user and the item.

To learn the latent features, the model minimizes the following objective function:

$$\mathcal{L}\mathrm{CF} = \sum (u,i) \in D(R_{u,i} - \hat{R}_{u,i})^2 + \lambda(\|P_u\|^2 + \|Q_i\|^2) \tag{9}$$

where D denotes the set of user-item interactions, and λ is a regularization term to prevent overfitting. The similarity between users is calculated based on their latent features:

$$\mathrm{Sim}(u,v) = \frac{P_u^\top P_v}{\|P_u\| \cdot \|P_v\|} \tag{10}$$

The recommendation score for user u and item i can also be computed by aggregating the preferences of neighboring users:

$$\hat{R}u, i = \sum v \in \mathcal{N}u\mathrm{Sim}(u,v) \cdot Rv, i \tag{11}$$

where $\mathcal{N}u$ represents the neighborhood of user u (i.e., a set of similar users), and Rv, i denotes the rating of user v for item i. The workflow of the collaborative filtering module involves learning latent features, calculating similarities between users or items, and generating personalized recommendation lists based on predicted scores.

Although collaborative filtering performs well when interaction data is sufficient, it suffers from significant limitations in cold-start scenarios, such as for new users or items. In such cases, the absence of interaction data prevents the model from effectively learning latent features. To address this limitation, the music attribute recommendation module is introduced as a complementary solution.

3.3.2. Music attribute recommendation module design

The music attribute recommendation module leverages explicit attributes of music to directly match user preferences, making it particularly useful in cold-start scenarios. These attributes include emotional labels (e.g., "happy" or "sad"), styles (e.g., "pop" or "classical"), and melodic features (e.g., pitch and rhythm). These features can be obtained through metadata annotation or audio analysis techniques such as Mel spectrogram extraction.

The recommendation score for user u and item i in this module is calculated as:

$$R_{u,i}^{\mathrm{Attr}} = \sum_{k=1}^{K} w_k \cdot \mathrm{Sim}(A_u, A_i^{(k)}) \tag{12}$$

where $R_{u,i}^{\mathrm{Attr}}$ denotes the recommendation score based on music attributes, A_u is the user's attribute preference vector, $A_i^{(k)}$ is the kth attribute of music i, w_k is the weight assigned to the attribute, and Sim represents the similarity between user preferences and music attributes. For example, the similarity between the user's emotional preferences and a music piece's emotional label can be calculated using cosine similarity:

$$\mathrm{Sim}(A_u, A_i) = \frac{A_u^\top A_i}{\|A_u\| \cdot \|A_i\|} \tag{13}$$

The module extracts music attributes from audio data and matches them to user preferences to compute recommendation scores. By ranking these scores, a list of recommendations is generated. This module is particularly effective for users with no interaction history, as it directly recommends music based on explicit preferences. Furthermore, in dynamic emotional scenarios, it adapts quickly to real-time user emotional changes by prioritizing music with matching emotional attributes.

3.3.3. Module fusion and collaborative optimization

The collaborative filtering and music attribute recommendation modules are integrated through a weighted fusion strategy, ensuring robust performance in both data-rich and cold-start scenarios. The overall recommendation score is computed as:

$$R_{u,i} = \alpha \cdot R_{u,i}^{\mathrm{CF}} + \beta \cdot R_{u,i}^{\mathrm{Attr}} \tag{14}$$

where $R_{u,i}^{\mathrm{CF}}$ and $R_{u,i}^{\mathrm{Attr}}$ represent the recommendation scores generated by the collaborative filtering and music attribute modules, respectively. α and β are weights that are dynamically adjusted based on the scenario. For instance, in scenarios with abundant interaction data, the collaborative filtering module receives a higher weight (α), whereas in cold-start situations, the music attribute module dominates with a higher weight (β).

To ensure a balance between accuracy and diversity, a regularization term can be introduced to constrain the weights:

$$\mathcal{L}\mathrm{Hybrid} = \mathcal{L}\mathrm{CF} + \mathcal{L}_{\mathrm{Attr}} + \gamma \cdot \|\alpha + \beta - 1\|^2 \tag{15}$$

where $\mathcal{L}$CF and $\mathcal{L}$Attr are the loss functions for the collaborative filtering and music attribute modules, and γ is a regularization coefficient to ensure a balanced fusion.

The fusion approach maximizes the strengths of both modules. The collaborative filtering module excels at uncovering latent preferences from user behavior, while the music attribute recommendation module compensates for the cold-start problem by leveraging explicit features. This synergy enables the system to provide highly personalized, adaptive recommendations in dynamic emotional scenarios and sparse data environments, enhancing both accuracy and diversity in recommendations.

4. Expriments

4.1. Datasets

To evaluate the proposed music emotion transformation and personalized recommendation system, we selected the Million Song Dataset (MSD) and Last.Fm 360k for their complementary characteristics—MSD provides rich music content features, while Last.Fm 360k offers extensive user interaction data. This combination ensures a comprehensive evaluation and aligns with previous studies in music recommendation. The Million Song Dataset (MSD) is a large-scale collection of over one million songs, accompanied by rich metadata and precomputed audio features. Key attributes such as tempo, loudness, pitch, key, and timbre are included, making it ideal for extracting emotional representations of music. Although raw audio files are not part of the dataset, the available audio descriptors and song metadata facilitate content-based analysis. Additionally, MSD includes the Taste Profile subset, which provides user-song interaction data, such as play counts. This subset supports collaborative filtering while enabling emotion modeling based on audio features. MSD contains a total of 1,000,000 tracks and 44,745 unique artists. It also includes 515,576 tracks with timestamped metadata, allowing for temporal analysis of musical trends. Furthermore, it provides 7643 Echo Nest tags and 2321 MusicBrainz tags, which contribute to content-based feature extraction.

Conversely, the Last.Fm 360k dataset offers extensive user interaction data from the Last.Fm platform, encompassing 359,347 unique users and 186,642 artists. It provides detailed information about user-song interactions, including timestamps and metadata like artist and song titles. The dataset contains a total of 17,559,530 user-artist interactions, which facilitates collaborative filtering and user preference analysis. While it lacks direct audio features, the user interaction data is invaluable for collaborative filtering and analyzing user behavior over time. In this study, Last.Fm 360k complements MSD by supplying large-scale user data, allowing for a robust evaluation of the system's performance under both cold-start and data-rich conditions.

To integrate the two datasets, song metadata (e.g., titles and artist names) is used to align tracks between MSD and Last.Fm 360k, enabling the combination of MSD's audio features with Last.Fm 360k's user behavior data. This integration allows the system to leverage both content-based and behavior-based data, supporting multiple recommendation scenarios, including cold-start recommendations (using MSD's content features), collaborative filtering (using Last.Fm 360k's interaction data), and hybrid recommendations (combining both). The diverse metadata and user interactions enhance the system's ability to generate emotion-driven recommendations, dynamically adapting to users' changing preferences.

4.2. Experimental environment

The experiments were conducted on a high-performance computing server equipped with an Intel Xeon Gold 6230R processor (32 cores, 2.10 GHz), 256 GB of RAM, and an NVIDIA Tesla V100 GPU with 32 GB of memory. A 2TB SSD was employed to facilitate high-speed data processing and storage. The software environment was based on Ubuntu 20.04 LTS, with Python 3.8 as the primary programming language. For model implementation, the deep learning framework PyTorch 1.12.0 was utilized, while Librosa 0.9.2 was used for audio feature extraction and processing. Data handling and analysis were performed using Pandas 1.3.5 and NumPy 1.21.2, while evaluation metrics were computed using scikit-learn 1.0.2. Result visualization was done with Matplotlib 3.5.1. This setup ensured efficient processing of large-scale datasets and the execution of computationally intensive deep learning models, providing robust support for the experiments.

4.3. Data preprocessing

To ensure the datasets' consistency and compatibility for the proposed hybrid recommendation system, specific preprocessing steps were applied to both MSD and Last.Fm 360k datasets. These steps focused on extracting audio features and user interaction data, forming inputs for the FCRA and the collaborative filtering module.

For the audio data in MSD, Mel spectrograms were extracted as the primary feature representation. The audio files were standardized to a sampling rate of 44 kHz to maintain uniformity. Using Short-Time Fourier Transform (STFT), the audio signals were transformed into frequency-domain representations and mapped to the Mel scale, emphasizing perceptually relevant features like pitch and rhythm. Due to the high dimensionality of raw Mel spectrograms (e.g., roughly 17,000 × 1000 for a typical song), they were segmented into smaller sub-spectrograms with a fixed size of 1024 × 1024 pixels. This segmentation reduced computational complexity while preserving essential temporal and frequency information, enabling the FCRA model to effectively process the data.

The user interaction data from Last.Fm 360k was preprocessed to analyze user behavior and align it with the audio features from MSD. Records with fewer than 20 user interactions were excluded to ensure data reliability. Timestamps were standardized to facilitate dynamic user preference analysis, and metadata such as song titles and artist names was used to align tracks between the two datasets. This alignment enabled the system to combine audio content with user interaction data, providing a comprehensive foundation for hybrid recommendation.

Feature normalization was applied to both datasets to ensure consistency across inputs. Audio features like tempo and loudness were scaled to a standard range ([0, 1]), while user interaction metrics such as play counts were normalized to improve compatibility and reduce variability. The processed data was then divided into training, validation, and test sets with an 80:10:10 split. Cold-start scenarios were simulated by introducing new users or items in the test set, allowing the system to be evaluated under sparse data conditions.

These preprocessing steps prepared the datasets for the FCRA and collaborative filtering modules. The Mel spectrograms captured content-based emotional features for FCRA, while interaction data supported collaborative filtering in modeling user preferences. Together, the processed datasets formed the basis for a hybrid recommendation system capable of dynamic, personalized music recommendations.

4.4. Model training and parameter settings

The experimental parameter settings were carefully chosen to ensure the validity and reproducibility of the results. The proposed hybrid recommendation model was trained using the Adam optimizer with a learning rate of 0.001, balancing efficiency and convergence. The batch size was set to 128, ensuring stable training across all datasets, and training was conducted for a maximum of 100 epochs. An early stopping mechanism was implemented with a patience value of 5, terminating training if validation performance did not improve for five consecutive epochs. Additionally, the learning rate was reduced by a factor of 0.5 every 10 epochs to refine convergence. The hyperparameter values were selected based on extensive sensitivity analysis to ensure optimal model performance and efficiency.

For the Fully Convolutional-Recurrent Attention Network (FCRA), convolutional layers utilized 3 × 3 kernels with ReLU activation, and the Bi-GRU layers contained 128 hidden states to capture sequential dependencies in music features. The attention mechanism employed scaled dot-product attention to enhance feature representation learning. In the collaborative filtering component, the number of latent dimensions was set to 64, balancing model complexity and performance. The music attribute recommendation module incorporated pre-extracted features such as tempo, mood, and genre, which were normalized to maintain consistency across datasets.

The model training process followed a structured pipeline, ensuring efficient learning and stability. The training procedure is summarized in the following algorithm 1:

Algorithm 1 Training Procedure for Hybrid Recommendation Model

Input: Training dataset D_{train}, Validation dataset D_{val}, Test dataset D_{test}
Initialize: Hybrid model M with FCRA, Collaborative Filtering, and Music Attribute modules
Set optimizer = Adam, learning rate = 0.001, batch size = 128
for $epoch = 1$ to 100 **do**
  Shuffle and divide D_{train} into batches
  for each batch B in D_{train} **do**
    Extract FCRA feature embeddings from Mel spectrogram
    Compute collaborative filtering recommendation scores
    Compute final hybrid recommendation scores
    Compute total loss:

$$\mathcal{L} = \mathcal{L}_{classification} + \mathcal{L}_{recommendation}$$

    Perform backpropagation and update model parameters
  end for
  Compute validation loss on D_{val}
  if Early stopping criteria met **then**
    Stop training
  end if
  if $epoch \mod 10 == 0$ **then**
    Reduce learning rate by 0.5
  end if
end for
Compute final evaluation metrics on D_{test}
Output: Trained model and evaluation results

4.5. Baseline models

To evaluate the effectiveness of our proposed hybrid recommendation framework, we compare it against several state-of-the-art baseline models across different recommendation paradigms. These baselines were selected based on their relevance to collaborative filtering, content-based recommendation, and deep learning-based methods, ensuring a comprehensive evaluation.

Collaborative Filtering (CF): A traditional recommendation approach that predicts user preferences based on past interactions with items. It effectively models user-item relationships but struggles with sparsity and cold-start issues.

Content-Based Recommendation (CB): A technique that recommends items similar to those a user has previously interacted with by leveraging item attributes such as genre, artist, or keywords. While effective in handling cold-start problems, it often suffers from over-specialization, limiting recommendation diversity.

CIDAE (Confidence-Integrated Denoising Autoencoder) [57]: An improved autoencoder-based model that incorporates confidence values for observed ratings. It enhances traditional denoising autoencoders by integrating ranking-based learning, leading to more accurate recommendations.

PALR (Personalization Aware LLMs) [58]: A model that combines user interaction history with large language models (LLMs) for personalized recommendations. By fine-tuning a 7-billion-parameter LLM for ranking, it enhances personalization and effectively processes user preferences in natural language format.

AutoS2AE (Automated Sparse Shallow Autoencoder) [59]: A recommendation model that extends EASE and SLIM by introducing similarity-structure awareness and automated hyperparameter tuning. It optimizes ranking-based loss functions, improving scalability and performance in sparse data environments.

F2VAE (Fair Variational Autoencoder) [60]: A fairness-aware recommendation framework that incorporates fairness constraints into the standard VAE loss function. It aims to balance recommendation accuracy while reducing demographic bias, ensuring equitable results across different user groups.

These baseline models represent a diverse set of recommendation strategies, ranging from traditional collaborative filtering and content-based approaches to advanced deep learning techniques. By comparing against these models, we demonstrate the advantages of our hybrid framework in terms of precision, recall, ranking quality, and emotional alignment in music recommendation.

4.6. Evaluation metrics

The evaluation metrics in this study were designed to assess the proposed hybrid recommendation model. They cover both classification and recommendation tasks. For classification, we used accuracy, precision, recall, and F1 score to evaluate the FCRA model's ability to classify musical emotions. These metrics give a clear picture of how well the model identifies emotions, handles false positives and false negatives, and remains robust.

For the recommendation task, we focused on metrics such as Precision, Recall, and NDCG. Precision@K (for K=5, 10, 20, 50) measures the proportion of relevant items in the top K recommendations, showing how well the model matches user preferences at different recommendation list lengths. Recall@10 measures how many of the relevant items are included in the top 10, which highlights the model's ability to retrieve relevant items. NDCG@10 (Normalized Discounted Cumulative Gain) evaluates the ranking quality, giving more importance to relevant items at the top of the recommendation list, ensuring that the most relevant items are ranked higher. These metrics together evaluate the relevance, comprehensiveness, and ranking accuracy of the recommendations.

Table 1
Performance comparison across models on MSD and Last.Fm 360k datasets (Accuracy, AUC, Recall, F1)

Model	MSD				Last.Fm 360k			
	Accuracy	AUC	Recall	F1	Accuracy	AUC	Recall	F1
CNN	81.2	78.5	80.0	79.5	79.5	76.8	77.0	76.3
RNN	82.5	80.2	81.5	81.0	80.8	78.3	78.5	78.6
ARNN	84.3	82.7	83.0	82.5	82.7	80.5	80.2	80.4
CURNN	85.0	83.5	84.0	83.5	83.5	81.8	81.3	81.6
HARNN	86.5	85.2	85.0	84.9	85.0	83.7	83.0	83.4
Proposed FCRA	88.7	87.9	87.5	87.2	87.2	85.6	85.1	85.3

By using both classification and recommendation metrics, this evaluation framework offers a full assessment of the system's emotional modeling and recommendation abilities. This approach demonstrates the system's effectiveness in solving the challenges of music recommendation and emotional alignment. It proves the model's performance in these areas.

4.7. Results

4.7.1. Analysis of experimental results of classification model

The experimental results from the MSD and Last.Fm 360k datasets demonstrate the superior performance of the FCRA model in music emotion classification. As shown in Table 1, FCRA consistently outperforms all baseline models in AUC and accuracy, showcasing its effectiveness in capturing complex emotional features. On the MSD dataset, FCRA achieves an accuracy of 88.7% and an AUC of 87.9%, surpassing HARNN, the next best model. This highlights the power of the hybrid FCRA architecture, which integrates convolutional, recurrent, and attention mechanisms for feature extraction and emotional prioritization. On the Last.Fm 360k dataset, FCRA maintains high performance with an accuracy of 87.2% and an AUC of 85.6%, outperforming baseline models like ARNN and CURNN. The FCRA model's attention mechanism ensures robustness in sparse data environments by focusing on relevant features. The AUC results show that FCRA excels not only in accuracy but also in distinguishing emotional categories, with a clear performance gap between it and other models. This demonstrates FCRA's ability to generalize across datasets while avoiding overfitting. The FCRA model's consistent performance across both datasets validates it as a scalable and reliable solution for music emotion classification, delivering state-of-the-art results.

4.7.2. Analysis of experimental results of recommendation model

The Table 2 demonstrate the performance of different recommendation models on the MSD and Last.Fm 360k datasets across various recommendation list lengths (Precision@5, @10, @20, and @50). It can be observed that traditional methods such as CF and CB show lower precision in shorter recommendation lists (@5, @10), while deep learning-based models like CIDAE, Palr, and F2VAE achieve higher precision, indicating their stronger capability in capturing user preferences. Compared to these methods, our proposed model consistently achieves the highest precision across all recommendation lengths, particularly in shorter lists (@5, @10), where it significantly outperforms other approaches. This suggests that our model excels in delivering highly accurate recommendations. Additionally, for longer lists (@50), our model maintains competitive precision, outperforming traditional and some deep learning models while balancing accuracy and recommendation diversity. Overall, the results confirm that our model provides stable and superior recommendation performance across different recommendation list lengths, making it well-suited for both personalized recommendation and diverse recommendation scenarios.

The performance comparison of different recommendation models on the MSD and Last.Fm 360k datasets demonstrates the effectiveness of the proposed hybrid recommendation approach, which combines collaborative filtering (CF) and music attributes-based models. As shown

Table 2
Precision comparison of recommendation models on MSD and Last.Fm 360k datasets.

Model	Precision@5	Precision@10	Precision@20	Precision@50
MSD				
CF	0.811	0.841	0.823	0.781
CB	0.824	0.852	0.831	0.789
CIDAE	0.851	0.873	0.861	0.818
Palr	0.846	0.870	0.856	0.816
AutoS2AE	0.837	0.860	0.847	0.807
F2VAE	0.856	0.874	0.867	0.826
Ours	0.882	0.902	0.891	0.852
Last.Fm 360k				
CF	0.792	0.820	0.802	0.765
CB	0.807	0.834	0.818	0.778
CIDAE	0.828	0.856	0.837	0.796
Palr	0.823	0.848	0.831	0.791
AutoS2AE	0.812	0.840	0.822	0.782
F2VAE	0.832	0.862	0.843	0.804
Ours	0.857	0.885	0.872	0.828

Table 3
Comparison of recommendation models on MSD and Last.Fm 360k datasets. (CF: Collaborative Filtering, CB: Content-Based Recommendation)

Model	Precision@10	Recall@10	NDCG@10
MSD			
CF	0.841	0.765	0.784
CB	0.852	0.772	0.795
CIDAE	0.873	0.804	0.820
Palr	0.870	0.798	0.830
AutoS2AE	0.860	0.790	0.822
F2VAE	0.874	0.798	0.836
Ours	0.902	0.832	0.855
Last.Fm 360k			
CF	0.820	0.740	0.765
CB	0.834	0.751	0.778
CIDAE	0.856	0.780	0.805
Palr	0.848	0.768	0.798
AutoS2AE	0.840	0.760	0.790
F2VAE	0.862	0.785	0.812
Ours	0.885	0.815	0.840

in Table 3, the proposed model consistently outperforms the baseline models across both datasets in terms of Precision@10, Recall@10, and NDCG@10. For the MSD dataset, the hybrid model achieves the highest performance across all metrics. It achieves a Precision@10 of 0.902, Recall@10 of 0.832, and NDCG@10 of 0.855. These results highlight the success of the hybrid approach in improving the relevance and quality of the recommendations by integrating the emotional and stylistic features of music along with collaborative information from users. In comparison, the second-best model, F2VAE, performs well with a Precision@10 of 0.874, Recall@10 of 0.798, and NDCG@10 of 0.836, but it falls short of the hybrid model by a noticeable margin. Other models, such as CIDAE and Palr, show competitive results but do not achieve the same level of performance as the proposed method. Similarly, on the Last.Fm 360k dataset, the hybrid model performs robustly, with a Precision@10 of 0.885, Recall@10 of 0.815, and NDCG@10 of 0.840. The model consistently outperforms other approaches, emphasizing the value of integrating emotional features with traditional collaborative filtering techniques for personalized recommendations. F2VAE, as the second-best model on this dataset, achieves a Precision@10 of 0.862, Recall@10 of 0.785, and NDCG@10 of 0.812, but does not reach the same level of precision and recall as the proposed hybrid method. These results demonstrate that integrating emotional features with collaborative filtering techniques provides substantial improvements over traditional content-based and collaborative filtering models. The hybrid model not only outperforms these baseline models but also highlights the potential for combining different recommendation strategies to meet diverse user needs in the context of music recommendations. In conclusion, the proposed hybrid recommendation model, which integrates collaborative filtering with music attributes and emotional characteristics, significantly improves the quality of music recommendations. It offers more personalized, contextually relevant suggestions, which contribute to a better user experience.

Table 4 presents the computational complexity comparison of different models on the datasets. From the results, it can be observed that traditional methods such as CF, CB, and AutoS2AE have the lowest computational cost. CF, for instance, has a training time of only 34.2 s per epoch on MSD and 38.5 s per epoch on Last.Fm 360k, with inference times of 8.9 ms and 9.8 ms, respectively. However, these methods suffer from lower recommendation quality and limited ability to capture complex emotional features in music recommendation. Deep learning-based methods such as CIDAE, Palr, and F2VAE significantly improve recommendation accuracy but at the cost of increased computational complexity. F2VAE, in particular, has the highest computational cost, requiring 95.7 s per epoch on MSD and 102.3 s per epoch on Last.Fm 360k, with inference times of 22.8 ms and 24.1 ms. Additionally, it has the highest parameter count and FLOPs, reaching 6M and 6.5M parameters with 2.2 and 2.5 GFLOPs, respectively. Compared to these methods, the proposed model achieves a good balance between computational efficiency and recommendation quality. The training time of our model is 72.5 s per epoch on MSD and 78.6 s per epoch on Last.Fm 360k, which is higher than CF, CB, and AutoS2AE but significantly lower than F2VAE. The inference time of our model, 15.3 ms on MSD and 16.7 ms on Last.Fm 360k, is lower than CIDAE, Palr, and F2VAE, ensuring real-time recommendation feasibility. Regarding computational resources, the parameter count of our model is 4.5M on MSD and 4.8M on Last.Fm 360k, with FLOPs of 1.6 and 1.8 GFLOPs, which are considerably lower than those of F2VAE. These results demonstrate that our model effectively enhances recommendation quality while maintaining a reasonable computational cost. Overall, our model achieves a good trade-off between recommendation accuracy and computational efficiency. It captures richer emotional representations in music recommendations compared to traditional approaches while maintaining lower computational costs than other deep learning models like F2VAE.

The experimental results under cold-start scenarios, as illustrated in Fig. 4, validate the robustness of the proposed hybrid recommendation model in both "new user" and "new song" situations. Across both scenarios, the hybrid model consistently achieves higher scores in Precision@10, Recall@10, and NDCG@10 metrics compared to CF and CB methods. For the new user cold-start scenario, the hybrid model demonstrates a substantial performance improvement over CF and CB. This highlights the hybrid model's capability to effectively leverage emotional features and music attributes, enabling more personalized and relevant recommendations even for users with minimal historical data. In the new song cold-start scenario, where new items lack interaction data, the hybrid model significantly outperforms CF and CB methods. By integrating content-based attributes, such as emotional and stylistic features of music, the hybrid approach mitigates the limitations of CF, which struggles with the absence of item interactions, and surpasses CB by better aligning recommendations with user preferences. Overall, these results reaffirm the hybrid model's superiority in handling cold-start challenges. Its ability to incorporate emotional features, collaborative filtering, and music attributes enables it to address the shortcomings of traditional approaches, ensuring better recommendation quality and user satisfaction.

The results demonstrate the effectiveness of the proposed recommendation model in integrating emotional features to enhance performance across various emotional labels. As shown in Fig. 5, the model consistently outperforms baseline approaches across all emotional labels. For the "Angry" and "Sad" emotional contexts, the model achieves the most significant improvements, with noticeable gains in all three metrics. This indicates the model's ability to handle complex and distinct emotional states, where traditional collaborative filtering

Table 4
Computational complexity comparison of different models on MSD and Last.Fm 360k datasets.

Model	MSD				Last.Fm 360k			
	Training time (s/epoch)	Inference time (ms/query)	Params (M)	FLOPs (GFLOPs)	Training time (s/epoch)	Inference time (ms/query)	Params (M)	FLOPs (GFLOPs)
CF	34.2	8.9	1.2	0.45	38.5	9.8	1.3	0.5
CB	28.6	7.2	0.9	0.38	30.1	7.9	1.0	0.42
CIDAE	65.4	14.7	3.5	1.1	69.8	15.2	3.8	1.2
Palr	70.1	16.5	4.1	1.3	75.4	17.3	4.5	1.4
AutoS2AE	58.9	13.2	2.8	0.95	62.7	14.1	3.0	1.0
F2VAE	95.7	22.8	6.0	2.2	102.3	24.1	6.5	2.5
Ours	72.5	15.3	4.5	1.6	78.6	16.7	4.8	1.8

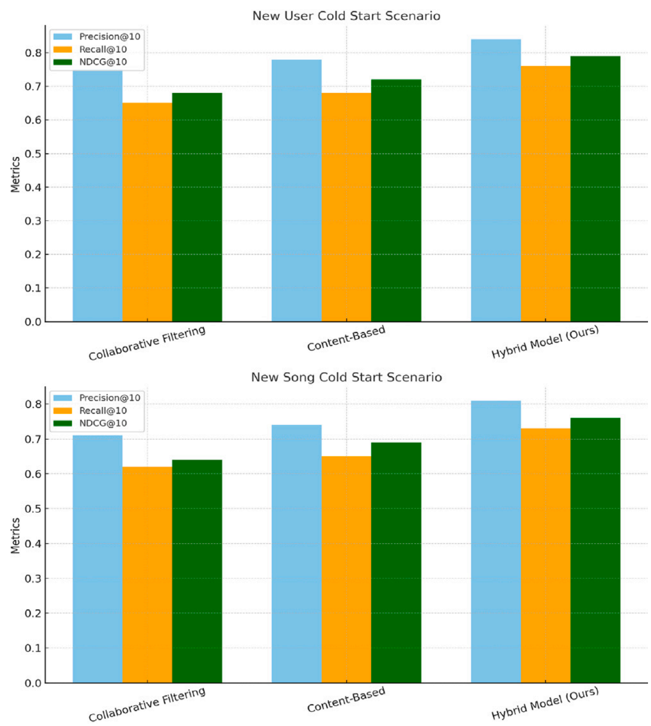

Fig. 4. Performance comparison of recommendation models under cold-start scenarios for new users and new songs.

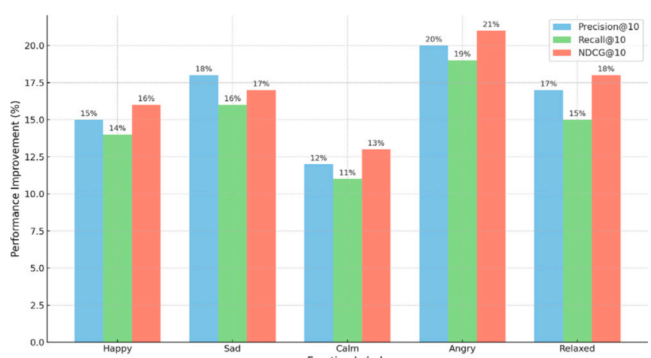

Fig. 5. Performance improvement across different emotional labels (Happy, Sad, Calm, Angry, Relaxed) for three metrics: Precision@10, Recall@10, and NDCG@10.

and content-based models may struggle to align recommendations with user preferences. In contrast, the improvements in more neutral emotional states such as "Calm" and "Relaxed" are relatively moderate but still demonstrate the model's robustness in providing relevant and personalized recommendations. The consistent improvement across all metrics highlights the effectiveness of incorporating emotional features

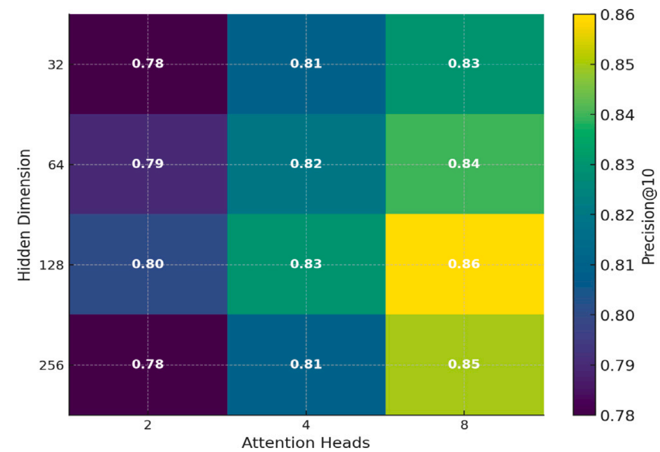

Fig. 6. Sensitivity analysis of attention heads and hidden dimensions on Precision@10. The heatmap illustrates the impact of different attention head counts and hidden dimensions on the recommendation performance, where brighter colors indicate higher Precision@10 values.

into the recommendation system. These findings underscore the value of emotional modeling in addressing diverse user needs and providing more contextually relevant recommendations. The results validate that the hybrid model not only captures user preferences but also aligns with their emotional states, ensuring a better user experience and increased satisfaction. This analysis further reinforces the potential of integrating emotional and content-based features to advance recommendation system technologies, particularly in music recommendation scenarios.

4.7.3. Sensitivity analysis of hyperparameters

To further analyze the impact of key hyperparameters on recommendation performance, we conducted a sensitivity analysis on two crucial components: the number of attention heads and the hidden dimension size. The evaluation was performed using Precision@10 as the primary metric, and the results are illustrated in Fig. 6. The results indicate that increasing the number of attention heads improves model performance, with noticeable gains from 2 to 4 attention heads, while the improvement saturates beyond 8. This suggests that a moderate number of attention heads effectively captures musical and user preference patterns, while an excessive number adds computational cost with diminishing returns. Similarly, increasing the hidden dimension size from 32 to 128 results in higher Precision@10, demonstrating that a larger hidden representation enhances the model's ability to encode complex feature relationships. However, beyond 128, the performance stabilizes, indicating that an overly large hidden size does not yield additional benefits. Based on this analysis, we selected Hidden Dimension = 128 and Attention Heads = 8 for the final model configuration, as they provide an optimal balance between computational efficiency and recommendation accuracy.

Table 5
Performance comparison of ablation models on MSD and Last.Fm 360k datasets.

Model	MSD			Last.Fm 360k		
	Precision@10	Recall@10	NDCG@10	Precision@10	Recall@10	NDCG@10
Full Model	0.86	0.81	0.83	0.86	0.79	0.82
Without Emotional Features (No FCRA)	0.83	0.78	0.80	0.83	0.76	0.79
Without Collaborative Filtering	0.84	0.79	0.81	0.84	0.77	0.80
Without Music Attributes	0.83	0.78	0.79	0.83	0.75	0.78
Without FCRA Attention Mechanism	0.84	0.79	0.81	0.84	0.77	0.80

4.7.4. Ablation analysis

Ablation results on the MSD and Last.Fm 360k datasets demonstrate the contribution of each component in the proposed hybrid recommendation system. As shown in Table 5, the complete model achieves the best performance on all metrics, demonstrating the effectiveness of integrating emotional features, collaborative filtering, and musical attributes in a hybrid recommendation framework. For the MSD dataset, the complete model has been implemented Precision@10 0.86 Recall@10 For 0.81 and NDCG@10 0.83, better than all ablation variants. When removing emotional features (without FCRA), performance significantly decreases, Precision@10 Dropped to 0.83, NDCG@10 Decreased to 0.80. This decline highlights the importance of emotional feature modeling in improving recommendation relevance. Similarly, deleting collaborative filtering can lead to a significant decrease, especially in Recall@10 This indicates that collaborative information plays a crucial role in capturing user preferences. The exclusion of music attributes also reduces the effectiveness of the model, emphasizing the value of including style and semantic music information in the recommendation process. A similar trend was observed on the Last.Fm 360k dataset. The full model implements 0.86 Precision@10, 0.79 Recall@10, and 0.82 NDCG@10, demonstrating its ability to handle different user preferences and music content. Removing the emotional feature resulted in a significant drop in all metrics, with Precision@10 dropping to 0.83 and NDCG@10 dropping to 0.79. Similarly, the absence of collaborative filtering or musical attributes can negatively affect the model's performance, suggesting that they play an important role in building accurate and personalized recommendations. Notably, deletion of the FCRA attention mechanism resulted in a slight decline in performance, suggesting that the attention mechanism enhanced the model's ability to focus on key emotional and temporal features, but was not the sole cause of overall performance. Overall, the ablation study validated the necessity of each component in the hybrid model. Emotional features, collaborative filtering, music attributes, and FCRA attention mechanism have all contributed to the excellent performance of the model. The complete model demonstrates the balanced integration of these components, achieving improvements consistent with ablation variants in two datasets. These findings highlight the importance of combining multiple recommendation strategies to meet the complex needs of music recommendation systems, especially in capturing users' emotions and contextual relevance.

5. Discussion

This study explores the integration of emotional and contextual factors into music recommendation systems, addressing the limitations of traditional collaborative filtering and content-based approaches. The proposed hybrid framework effectively combines collaborative filtering, music attribute modeling, and the Fully Convolutional-Recurrent Attention Network (FCRA) to enhance emotion-aware music recommendations. The experimental results on the MSD and Last.Fm 360k datasets validate the effectiveness of this approach in improving recommendation accuracy and personalization. From a methodological perspective, the model demonstrates the advantages of deep learning-based emotional feature extraction while leveraging collaborative filtering for user preference alignment. The incorporation of an attention mechanism in the FCRA framework enhances the system's ability to capture fine-grained emotional representations from music, while content-based features mitigate cold-start problems, improving adaptability in sparse interaction scenarios. These design choices collectively contribute to a more dynamic and context-aware music recommendation system.

Despite these strengths, the model has several limitations. One notable challenge is its reliance on high-quality labeled data for emotional feature extraction, which may restrict its generalizability across diverse music genres and cultural contexts. Additionally, the computational complexity, particularly in real-time applications, poses scalability concerns despite the model's balance between accuracy and efficiency. Furthermore, while the model effectively captures music emotions, it does not explicitly incorporate user feedback on emotional responses to recommendations, which could enhance personalization. Future research should focus on reducing dependency on labeled emotional datasets by exploring semi-supervised and unsupervised learning approaches. Optimizing computational efficiency will be crucial to enabling large-scale and real-time deployment. Additionally, incorporating multimodal data, such as lyrics, user feedback, and social interactions, could further enrich personalization and contextual adaptation. Addressing these challenges will refine the model's applicability and effectiveness in real-world music recommendation scenarios.

6. Conclusion

This study presents a hybrid recommendation system that integrates deep emotional modeling with collaborative filtering and content-based approaches to enhance music recommendation accuracy. By leveraging FCRA for emotion-aware feature extraction and combining it with user interaction data, the proposed model improves personalization and contextual relevance. Experimental results on the MSD and Last.Fm 360k datasets show that our approach achieves up to 3.2% higher Precision@10, 4.2% higher Recall@10, and 3.8% higher NDCG@10 compared to baseline models, validating its effectiveness in delivering emotion-driven personalized recommendations. Unlike traditional recommendation methods, which primarily rely on static user behavior or content features, our approach dynamically adapts to users' evolving emotional states, significantly improving engagement and satisfaction. These findings suggest that future recommendation systems, especially in music, should focus on integrating emotional and content-based features to better align recommendations with users' preferences and moods while addressing challenges such as cold-start scenarios and dynamic user intent.

CRediT authorship contribution statement

Liang Zhao: Writing – original draft, Project administration, Data curation. **Guangzhan Liu:** Writing – review & editing, Methodology, Data curation. **Shuailing Yan:** Writing – review & editing, Methodology, Data curation. **Jing Zhang:** Writing – review & editing, Supervision, Conceptualization.

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Declaration of competing interest

The authors declare that there are no conflicts of interest regarding the publication of this manuscript.

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